# MANJUSHA ROY CHOUDHURY

R&D DATA SCIENTIST (O-1A & EB-1A I140 APPROVED)

## CONTACT

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Santa Clara, CA 95050

Manjusha - LinkedIn

## SKILLS

Python | SciPy | SQL | Java | Amazon Web Services | BEAST Epidemiology | Pandas | Numpy | Git | R | Apache Spark | Hadoop

## PUBLICATIONS

- Published in high-impact journals: Nature Chemistry, Gynecologic Oncology.
- Manjusha Roy Choudhury - Google Scholar

## PATENT

US Patent App #17757644 (Apr 2023)
Wang, B.; Boykin. D.W.; Roy Choudhury, M.A.; Kumar, A.; Yu, B.; Zhu, M.
"Compounds for the treatment of bacterial infection and potentiation of antibiotics."

## TALKS

**AI in Heathcare & Pharma Summit 2024**

Harnessing AI for the Development of a Blood-Based Diagnostic Test for Adnexal Mass Risk Assessment

## PROFILE

A human diseases researcher, exploring new drugs and healthcare outcomes. Specializing in AI and ML applications within the health domain.

## WORK EXPERIENCE

**Aspira Women's Health** June 2023 - Present

R&D Data Scientist

- Led the development and implementation of an AI reliability and explainability pipeline and dashboard, enhancing model transparency and trustworthiness in decision-making processes.
- Contributed to the discovery of tests that led to a 62% reduction in incorrect surgical diagnoses for ovarian cancer (published: Frontiers in Medicine).
- Developed a novel ML diagnostic model for longitudinal monitoring of ovarian cancer risk in women with indeterminate or low-risk adnexal masses (published: Gynecologic Oncology).
- Designed ML models for early detection of endometriosis using biomarker and patient data, achieving an ROC AUC of 82.

**Prescryptive Health Inc.** May 2022 - May 2023

AI Data Scientist

- Developed predictive drug pricing models for pharmacies, resulting in a 37% uplift in gross margins for both cold-start and warm-start customers.
- Trained random-forest models on historical claims data for pricing different purchase quantities.
- Designed visualizations and analyzed pricing data to explore trade-offs of different pricing models with customers.
- Built data pipelines to manage large volumes of semi-structured data from multiple customers.

## EDUCATION

**Ph.D. Chemistry, Masters Data Science** 2015 - 2021
Georgia State University
**GPA:** 3.96 / 4.0

**Masters Chemistry** 2011 - 2013
University of Manchester
**GPA:** Research Course

**Bachelors Chemistry** 2008 - 2011
St. Stephen's College, Delhi University
**GPA:** 3.8 / 4.0

## INTERNSHIPS

Biocon, Bangalore – Process Development Lab
Ranbaxy, New Delhi – Materials Chemistry Lab
GE Global Research, Bangalore – Materials Chemistry Lab
Indian Institute of Technology, Kanpur – Analytical Chemistry Lab